



Software Relevant Tools, Standards, and Code Versioning using Github

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Program:	CpE	Section:	B3

A. Learning Outcomes:

At the end of this laboratory, students should be able to:

- 1. Identify software development tools and standards used in software projects.
- 2. Create and manage a GitHub repository.
- 3. Apply Git version control commands.
- 4. Collaborate using GitHub features.
- 5. Demonstrate proper documentation and coding standards.

B. Laboratory Scenario:

You have been hired as a junior software developer in a software company. Your team uses GitHub for source code management and follows coding standards to maintain software quality. Your task is to create a project repository, manage versions of source code, and document your work according to industry standards.

C. Practical Tasks:

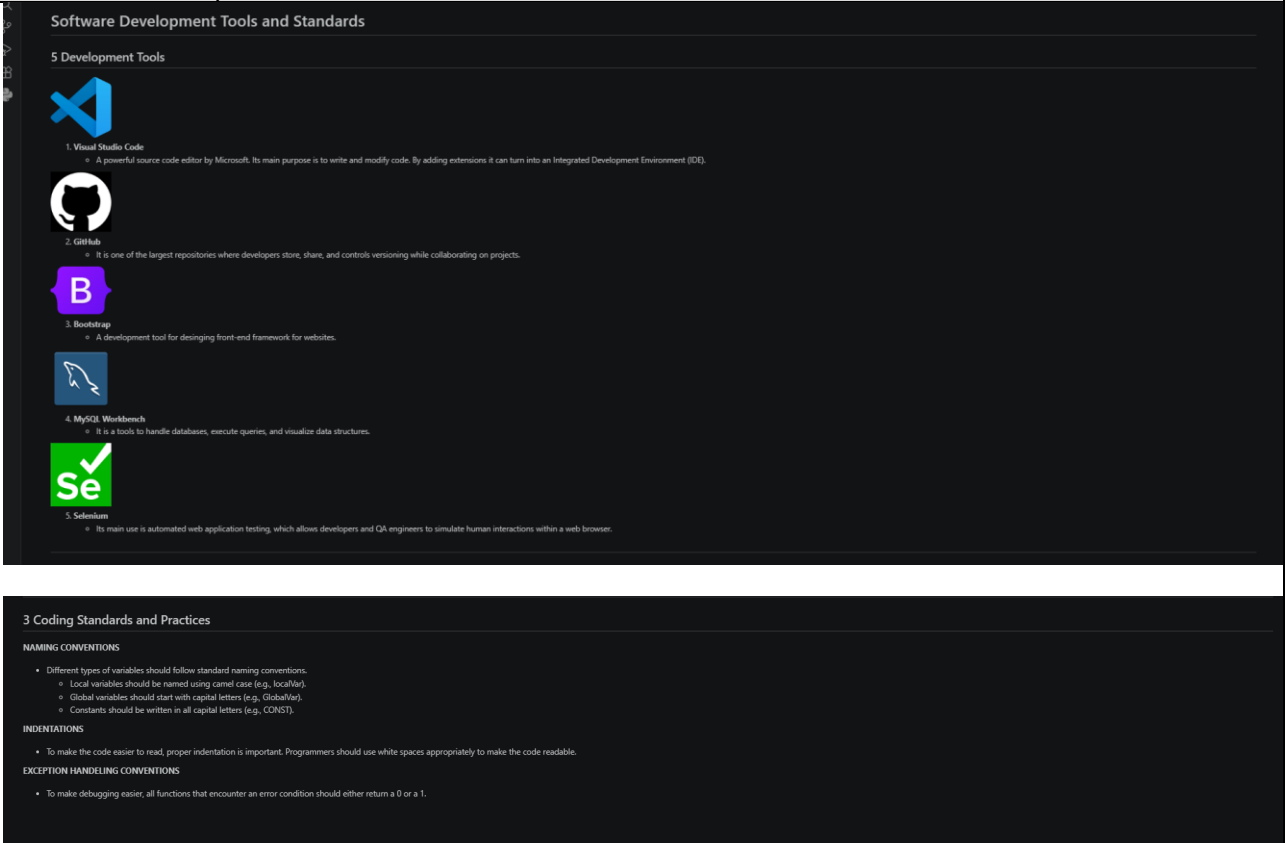
1. Software Development Tools and Standards

Create a document named Tools_and_Standards.md containing:

- a. List at least 5 development tools.
- b. Describe the purpose of each tool.
- c. Identify at least three coding standards or best practices used in software development.

(Please include the following: 1. Screenshots of the output)

Screenshot of Output



*Resize the image if necessary.

2. GitHub Repository Creation

- a. Create a GitHub account (if none exists).
- b. Create a public repository named: -
- c. Add the following:
 - i. README.md
 - ii. LICENSE

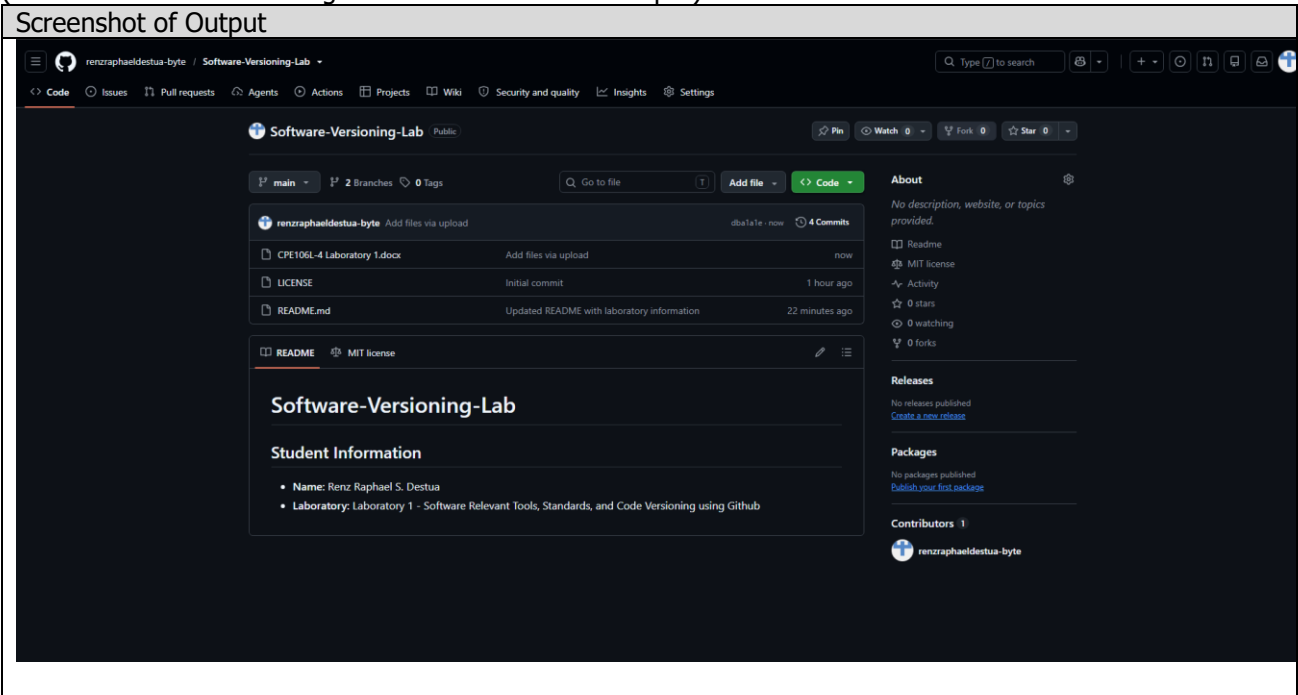


Laboratory Report 1

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- iii. .gitignore
- d. Upload this laboratory file in this repository once accomplished.
- e. Submit GitHub repository link
- f. Screenshot of repository homepage

(Please include the following: 1. Screenshots of the output)



- 3. Git Version Control
 - a. Perform the following Git commands and document them in a file named: (Git_Report.md)
 - i. Clone Repository
 - a) git clone <repository-url>
 - ii. Creating New Branch
 - a) git checkout -b feature-update
 - iii. Modify README.md
 - a) Add your name and laboratory information
 - iv. Commit Changes
 - a) git add .
 - b) git commit -m "Updated README with laboratory information"
 - v. Push Changes
 - a) git push origin feature-update
 - vi. Create Pull Request
 - a) feature-update → main
 - b. Screenshot the following:
 - i. Commands executed
 - ii. Pull Request
 - iii. Commit history



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(Please include the following: 1. Screenshots of the output)

Screenshot of Output

i. ii.

```
PS C:\Users\Renz\Software-Versioning-Lab> git checkout -b feature-update
Switched to a new branch 'feature-update'
PS C:\Users\Renz\Software-Versioning-Lab> git add .
PS C:\Users\Renz\Software-Versioning-Lab> git commit -m "Updated README with laboratory information"
Author identity unknown

*** Please tell me who you are.

Run

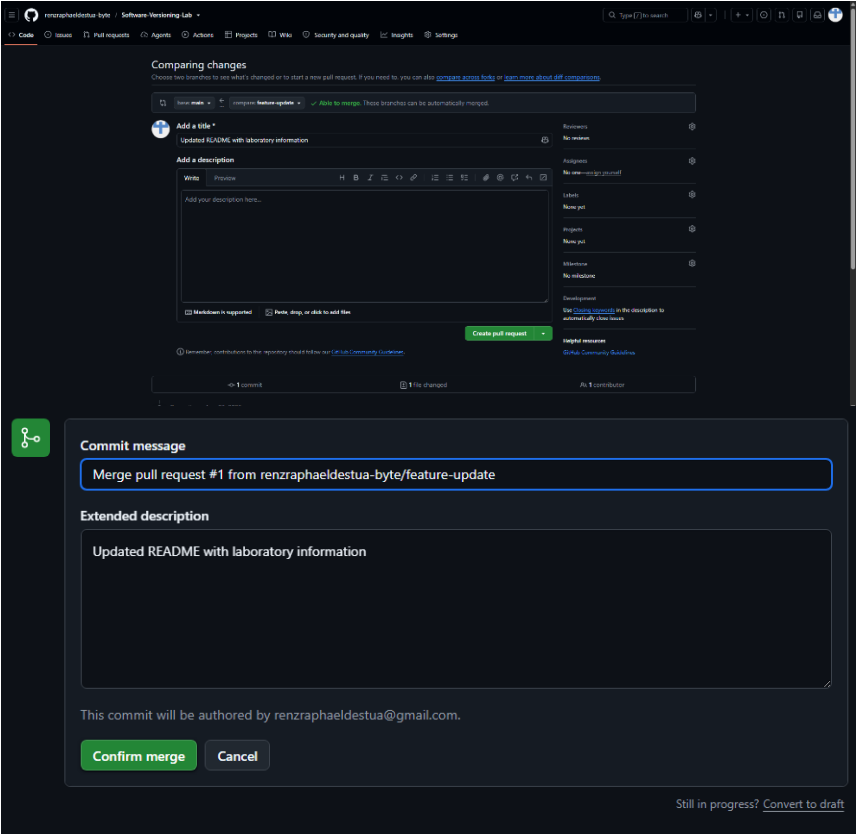
  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Renz@LAPTOP-IJBA0001.(none)')
PS C:\Users\Renz\Software-Versioning-Lab> git config --global user.email "renzraphaeldestua@gmail.com"
PS C:\Users\Renz\Software-Versioning-Lab> git push origin feature-update
info: please complete authentication in your browser...
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-update' on GitHub by visiting:
remote:   https://github.com/renzraphaeldestua-byte/Software-Versioning-Lab/pull/new/feature-update
remote:
To https://github.com/renzraphaeldestua-byte/Software-Versioning-Lab.git
 * [new branch]      feature-update -> feature-update
PS C:\Users\Renz\Software-Versioning-Lab> git status
On branch feature-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

PS C:\Users\Renz\Software-Versioning-Lab> git commit -m "Updated README with laboratory information"
[feature-update b91685b] Updated README with laboratory information
 1 file changed, 5 insertions(+), 1 deletion(-)
PS C:\Users\Renz\Software-Versioning-Lab> git push origin feature-update
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 445 bytes | 445.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/renzraphaeldestua-byte/Software-Versioning-Lab.git
 0eeeb1..b91685b feature-update -> feature-update
PS C:\Users\Renz\Software-Versioning-Lab> |
```

iii.



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D. Findings, Observations, and Comments:

Guide Questions:

- 1. What additional steps are taken in this activity?
- 2. Are there any difficulties during the GitHub setup?
- 3. What are your findings, observations, and comments in this exercise?

- 1. The additionally steps I've taken on this activity is the insertion of logo image in the 5 Development tools for better understanding and what does the tool looks like.
- 2. Since I have a surface level prior understanding of GitHub It was not difficult to set up my GitHub.
- 3. I was able to learn the different developer tools and what their purpose is. Additionally, I learned how to use GitHub locally using git and in PowerShell.

PICTURE/PROOF OF DOING THE ACTIVITY





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E. Score Sheet

Criteria	Poor (25%)	Fair (50%)	Good (75%)	Excellent (100%)	Score
I. Completeness, Documentation, and Organization of Report	The report was incomplete, messy, and had many unanswered questions.	The report was complete, messy, and had many unanswered questions.	The report was complete, neat, and had 1 or 2 unanswered questions.	The report was complete, neat, and all the questions had been answered.	
II. Coding Design and Patterns	The program had inappropriate elements and structures, incorrect indentation, and poor program documentation.	The program had corrected indentation, but inappropriate elements and structures and poor program documentation.	The program had corrected indentation, adequate program documentation, but inappropriate elements and structures.	The program had appropriate elements and structures, correct indentation, and excellent program documentation.	
III. Findings, Observations, and Comments	The findings, observations, and comments were not based on the gathered data and results. All thoughts were inconsistent and unclear.	The findings, observations, and comments based on the gathered data and results but were not elaborated. Not all thoughts were consistent and clear.	The findings, observations, and comments were based on the gathered data and results and were mostly elaborated. Most of the thoughts were consistent and clear.	The findings, observations, and comments were based on the gathered data and results and were completely elaborated. All the thoughts were consistent and clear.	
IV. Grammar and Composition	The words used were inappropriate, wrong grammar usage, and poor sentence construction was observed.	The words used were appropriate; however, bad grammar and poor sentence construction were observed.	The words used were appropriate, had proper grammar usage, but poor sentence construction was observed.	The words used were appropriate, had proper grammar usage, and excellent sentence construction.	
SCORE:					

CHECKED			
Rating		Signature	
		Date	